Layout of Standard Cells - DRC, LVS, PEX

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***Abstract*—This paper presents the layouts of NAND3 and NOR2 gates, as well as a mirror adder, along with delay simulations conducted before and after parasitic extraction. All designs were created and simulated using the gpdk45nm technology in the Cadence Design Suite.**

I. Introduction

Following constraints were taken into consideration:

TABLE I: DESIGN PARAMETERS

Parameter	Value
L_{min}	45 nm
W_{min}	120 nm
V_{DD}	1 V
Cell Height	2 µm
VDD/GND Rail Width	0.12 µm

II. Question 1

Create the schematic, symbol, and layout of NOR2 and NAND3 gates with appropriate sizing. Assume $L = L_{min}$ for all transistors, and W_{NMOS} = 260 nm, W_{PMOS} = 390 nm for a unit inverter. Measure $t_{pd} = (t_{pdf} + t_{pdr})/2$ with and without layout parasitics.

Solution

The layouts and corresponding testbench configurations for the NOR2 and NAND3 gates were designed, simulated, and analyzed to evaluate their timing performance. In the pre-layout simulations, the delays were found to be relatively lower as the models considered only idealized device parameters without interconnect effects. Post-layout simulations, performed after parasitic extraction, revealed significantly higher delays. This increase is primarily due to the impact of parasitic capacitances from the metal interconnects and contacts, as well as parasitic resistances in the routing paths. These parasitics introduce additional RC delay into the signal propagation, slowing down the gate transitions. Furthermore, the increased capacitive loading affects both the rise and fall times, leading to asymmetry in the delay characteristics. This highlights the importance of layout-aware timing analysis in accurately predicting real-world circuit performance.

TABLE II: Delay values obtained prior to and following parasitic extraction

Gate	t_{pd} (ps)	t_{pd} after PEX (ps)
NOR2	46.17	62.71
NAND3	49.33	70.19

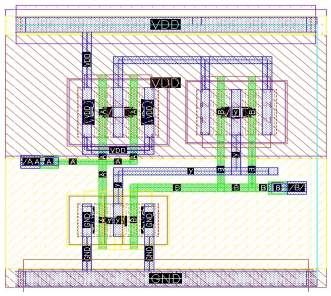


Fig. 1: NOR (2 input) Layout

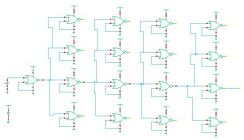


Fig. 2: NOR (2 input) Schematic

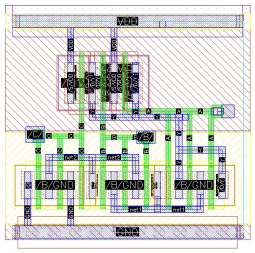


Fig. 3: NAND (3 input) Layout

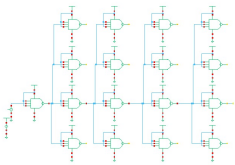


Fig. 4: NAND (3 input) Schematic

III. Question 2

A standard cell is characterized for various input transi tions times and output load capacitances (among many other variables) and provided as a liberty file. Characterize your NOR2 gate (post layout) and populate the following table for the worst-case (across input patterns) fall delay at the slowest

PVT corner

Solution

At the slowest PVT corner—125°C, SS process, and $V_{DD} = 0.9\,\text{V}$—a post-layout analysis of the NOR2 gate was performed for different load capacitances (ranging from 0.1 fF to 2 fF) and input transition times (ranging from 10 ps to 200 ps). The corresponding fall delay results are listed bewlow in the TABLE III.

TABLE III: LIBERTY FILE

Time (ps)	CL=0.1fF	CL=0.2fF	CL=0.3fF	CL=0.4fF	CL=0.5fF	CL=0.6fF	CL=0.7fF	CL=0.8fF	CL=0.9fF	CL=1fF	CL=1.1fF	CL=1.2fF	CL=1.3fF	CL=1.4fF	CL=1.5fF	CL=1.6fF	CL=1.7fF	CL=1.8fF	CL=1.9fF	CL=2fF
10	16.85	17.62	18.41	19.19	19.94	20.72	21.49	22.24	23.03	23.79	24.55	25.31	26.07	26.85	27.6	28.36	29.11	29.89	30.61	31.4
20	19.84	20.64	21.44	22.22	23	23.77	24.53	25.32	26.1	26.86	27.62	28.39	29.16	29.91	30.69	31.44	32.19	32.99	33.75	34.48
30	22.88	23.66	24.47	25.23	26.02	26.82	27.61	28.38	29.14	29.9	30.68	31.46	32.22	32.99	33.76	34.55	35.29	36.09	36.82	37.57
40	25.9	26.69	27.45	28.25	29.04	29.81	30.62	31.39	32.15	32.94	33.73	34.5	35.28	36.04	36.85	37.61	38.38	39.16	39.92	40.66
50	28.98	29.77	30.57	31.36	32.16	32.96	33.74	34.52	35.29	36.08	36.88	37.65	38.44	39.17	39.96	40.74	41.5	42.24	43	43.79
60	32.07	32.86	33.65	34.47	35.25	36.05	36.84	37.63	38.36	39.14	39.92	40.72	41.49	42.26	43.04	43.83	44.61	45.36	46.1	46.94
70	35.22	36.01	36.79	37.6	38.38	39.16	39.95	40.73	41.54	42.32	43.1	43.87	44.65	45.43	46.22	46.98	47.76	48.53	49.32	50.14
80	38.17	39.05	39.92	40.73	41.52	42.32	43.09	43.88	44.68	45.47	46.23	47.05	47.83	48.62	49.38	50.14	50.91	51.71	52.49	53.23
90	41.03	41.99	42.89	43.78	44.66	45.47	46.28	47.06	47.84	48.63	49.4	50.18	50.96	51.74	52.5	53.28	54.08	54.85	55.62	56.38
100	43.78	44.79	45.81	46.75	47.67	48.52	49.37	50.21	50.98	51.76	52.54	53.32	54.09	54.87	55.65	56.43	57.21	57.97	58.74	59.52
110	46.44	47.54	48.57	49.57	50.52	51.47	52.36	53.22	54.07	54.89	55.69	56.47	57.24	58.02	58.79	59.56	60.35	61.12	61.9	62.67
120	49	50.15	51.24	52.31	53.3	54.29	55.25	56.17	57.06	57.93	58.75	59.59	60.39	61.17	61.94	62.72	63.49	64.27	65.04	65.81
130	51.47	52.67	53.83	54.94	56.03	57.05	58.04	59.01	59.96	60.87	61.76	62.6	63.46	64.28	65.09	65.87	66.64	67.41	68.19	68.96
140	53.88	55.16	56.37	57.52	58.65	59.73	60.8	61.79	62.79	63.72	64.65	65.55	66.45	67.31	68.14	68.97	69.79	70.57	71.34	72.11
150	56.22	57.56	58.83	60.04	61.23	62.35	63.45	64.51	65.54	66.52	67.48	68.42	69.34	70.25	71.13	72	72.84	73.65	74.47	75.26
160	58.5	59.89	61.24	62.51	63.74	64.91	66.06	67.16	68.21	69.24	70.26	71.26	72.21	73.15	74.07	74.95	75.82	76.69	77.53	78.33
170	60.74	62.2	63.58	64.91	66.18	67.41	68.6	69.73	70.83	71.93	72.98	74	75	75.97	76.92	77.85	78.75	79.64	80.51	81.38
180	62.93	64.42	65.87	67.24	68.58	69.87	71.07	72.26	73.41	74.54	75.64	76.69	77.73	78.73	79.72	80.69	81.63	82.54	83.43	84.32
190	65.07	66.64	68.13	69.51	70.94	72.26	73.51	74.74	75.96	77.11	78.24	79.34	80.41	81.44	82.47	83.45	84.43	85.39	86.33	87.25
200	67.18	68.78	70.32	71.79	73.21	74.61	75.92	77.19	78.44	79.64	80.8	81.93	83.04	84.11	85.17	86.19	87.19	88.19	89.16	90.1

IV. Question 3

Create the schematic and layout of a mirror full-adder (FA) cell (Fig. 11.4 (c) in Weste-Harris Book). Characterize and summarize all interesting features post layout– including delay, area, power. Name this cell: FA_D22

Solution

The schematic and layout of full adder simuulated is shown as follows

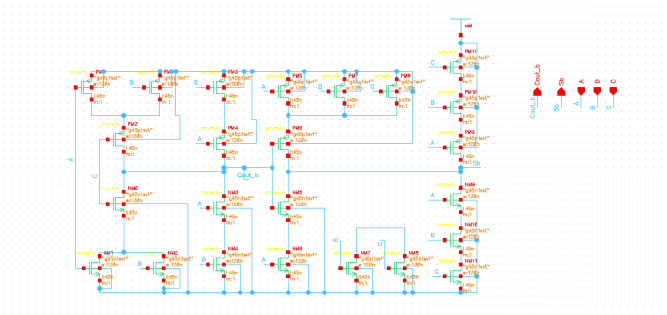


Fig. 5: Full Adder Schematic

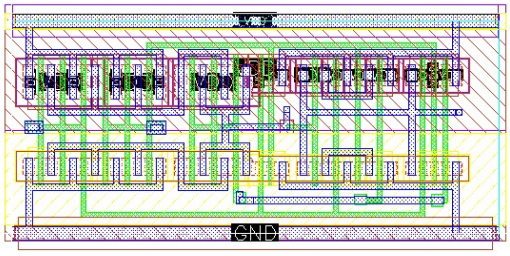


Fig. 6: Full Adder Layout

The parameters including SUM' delay, COUT' delay, power, and area are summarized in the table below. All calculations were performed after parasitic extraction, and the table shows that the SUM' delay is greater than the COUT' delay.

TABLE IV: Parameters simulated after PEX

PARAMETER	VALUE
SUM' delay	50.99 ps
COUT' delay	15.80ps
Area	9.2 μm^2
Power	1.789 μ W

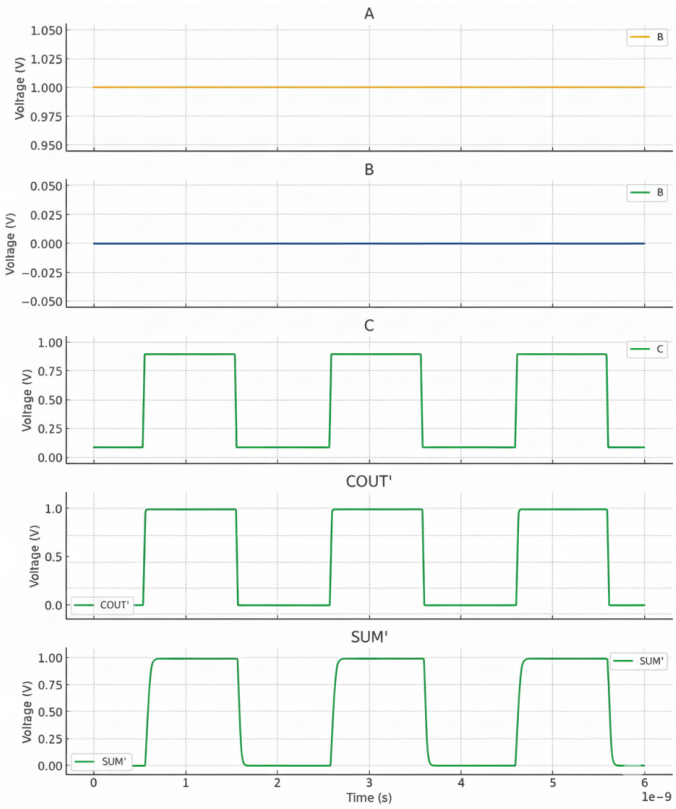


Fig. 7: Waveforms

V. Question 4

The layout of a Flip-flop (with set) conforming to the above specifications is available using the following steps. Create the schematic for this and verify that it works as expected (functional simulation). What is the $t_{clocktoq}$ delay of this FF before PEX and post-PEX?

Solution

The schematic extracted from the layout given below is matched with the layout and confirmed running LVS.

The delay measured before and after PEX is summerized in the below table all these have been done after matching the schematic with the layout provided to us using DRC and LVS checks

TABLE V: Delays simulated prior to and post PEX

Condition	Delay
PRE LAYOUT	76.3 ps
COUT' delay	140.80 ps

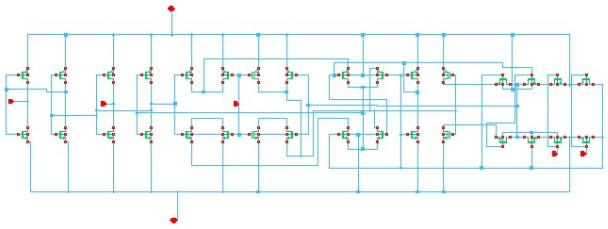


Fig. 8: Schematic of DFF

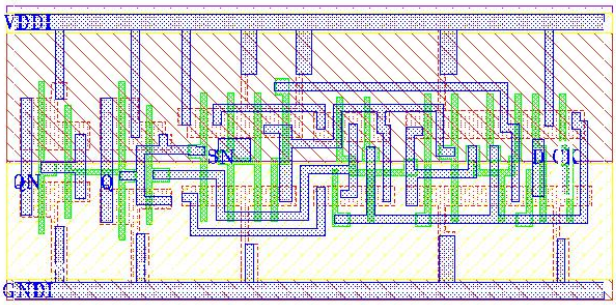


Fig. 9: Dff layout

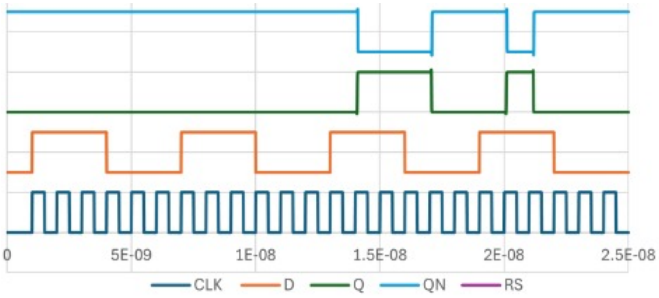


Fig. 10: SIGNALS

VI. Conclusion

Simulations of the schematic, layout, and post-layout (PEX) stages were successfully completed, providing a comprehensive verification of the design at multiple levels. It was observed that both propagation delay and power consumption increase after parasitic extraction compared to the schematic-level analysis. This is expected since parasitic capacitances and resistances introduced by the layout interconnects add extra loading to the circuit, slowing down signal transition and increasing power dissipation. Despite these increases, the extracted performance metrics remain within acceptable design limits, ensuring that the circuit meets specifications and operates reliably in real scenarios where parasitic effects are present. This highlights the importance of post-layout simulation for accurately predicting the real-world behavior of integrated circuits.

References

[1] N. H. E. Weste and D. M. Harris, *CMOS VLSI Design: A Circuits and Systems Perspective*, 4th ed., Addison Wesley, Pearson, 2011.

[2] Cadence Design Systems, *GPDK045 Reference Manual: 45nm Process Design Kit*, Revision 6.0, 2019.

[3] K. Sindhuja, "Design and Analysis of 4:2 Priority Encoder in 45nm Using Cadence GPDK Technology," *International Journal of Science, Engineering and Technology*, vol. 13, no. 3, 2025.

[4] F. Approximation, "Design and Simulation of Various Approximation Adders to Reduce the Power Consumption," *International Journal of VLSI System Design and Communication Systems*, vol. 2, no. 9, 2014.

[5] Cadence Community, "Regarding Cadence Virtuoso gpdk45nm technology," 2015. [Online].